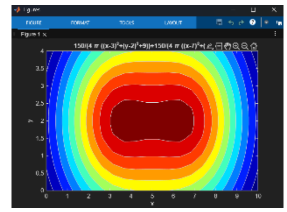
Two 150 Watt Bulbs

In this case we need to decide where to put the two bulbs. Common and sense tells us to arrange the bulbs symmetrically a long a line down the center of a room in the long direction; that is along the line y=2. Define a function that gives the intensity of light at a point (x, y) on the floor due to a 150 watt bulb at a position (d, 2) on the ceiling.

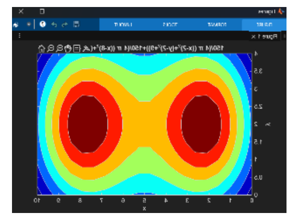
```
>> I(x, y) = 150/(4π((x-d)^2+(y-2)^2+9)
```

Let's get an idea of the illumination pattern if we put one light at d=3 and the other at d=7. We specify the drawings of 20 contours in this and the following plots



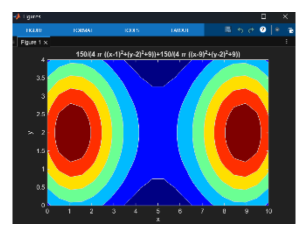
```
>> f = @(x,y) 150 ./ (4*pi*((x-3).^2 + (y-2).^2 + 9)) + 150 ./ (4*pi*((x-7).^2 + (y-2).^2 + 9));
>> ezcontourf(f, [0 10 0 4], 20);
>> colormap(jet);
```

The floor is more evenly lit with one bulb, it looks as if the bulbs are closer together as they should be. If we move the bulbs further apart, the center of the room will get dimmer, but the corners will get brighter. Let's try changing the location of the lights to d=2 and d=8.



```
>> f = @(x,y) 150 ./ (4*pi*((x-2).^2 + (y-2).^2 + 9)) + 150 ./ (4*pi*((x-8).^2 + (y-2).^2 + 9));
>> ezcontourf(f, [0 10 0 4], 20);
>> colormap(jet);
```

This is an improvement. The corners are the darkest spots of the room, though the light intensity along the walls toward the middle of the room near (x=5) is diminishing as we move the bulbs further apart. Still, to better illuminate the darkest spots we should keep moving the bulbs apart. Let's try lights at d=1 and d=9.



```
>> f = @(x,y) 150 ./ (4*pi*((x-1).^2 + (y-2).^2 + 9)) + 150 ./ (4*pi*((x-9).^2 + (y-2).^2 + 9));
>> ezcontourf(f, [0 10 0 4], 20);
>> colormap(jet);
```